

ST117 Individual DRAFT Written Report - Phase 2 Task 4 (Fine grains - Butterflies)

Report Pod MS-10

08-05-2026

a)

```
# Read Datasets
fineGrain <- read.csv("Vegetation/ECN_VF1.csv")
butterflies <- read.csv("Invertebrates/ECN_IB1.csv")

# Sort Butterflies by Year
butterflies$SDATE = as.Date(butterflies$SDATE, format = "%d-%b-%y")
butterflies$YEAR <- year(butterflies$SDATE)

# Sum all butterfly counts per site per year across all species and transect sections
butterfly_summary <- butterflies %>%
  group_by(SITECODE, YEAR) %>%
  summarise(
    Total_Butterflies = sum(VALUE, na.rm = TRUE),
    .groups = "drop"
  )

# Count unique species per site per year as a measure of plant diversity
fineGrain_summary <- fineGrain %>%
  filter(!is.na(VALUE)) %>%
  group_by(SITECODE, SYEAR) %>%
  summarise(
    Unique_FG_Species = n_distinct(FIELDNAME),
    .groups = "drop"
  ) %>%
  rename(YEAR = SYEAR)

# Constructed Overall Dataset
combined_dataset <- butterfly_summary %>%
  left_join(fineGrain_summary, by = c("SITECODE", "YEAR"))

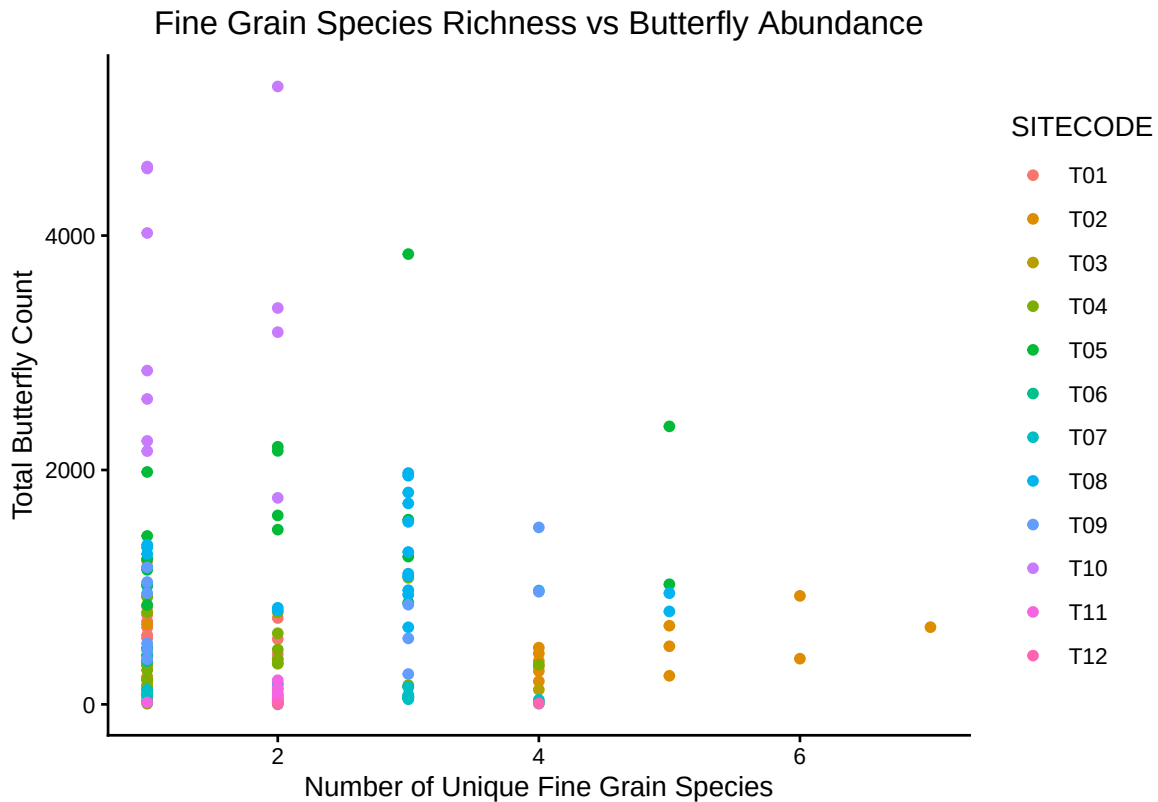
kable(head(combined_dataset, 3), caption = "First Three Rows of Combined Dataset")
```

Table 1: First Three Rows of Combined Dataset

SITECODE	YEAR	Total_Butterflies	Unique_FG_Species
T01	1993	302	NA
T01	1994	428	2
T01	1995	635	NA

b)

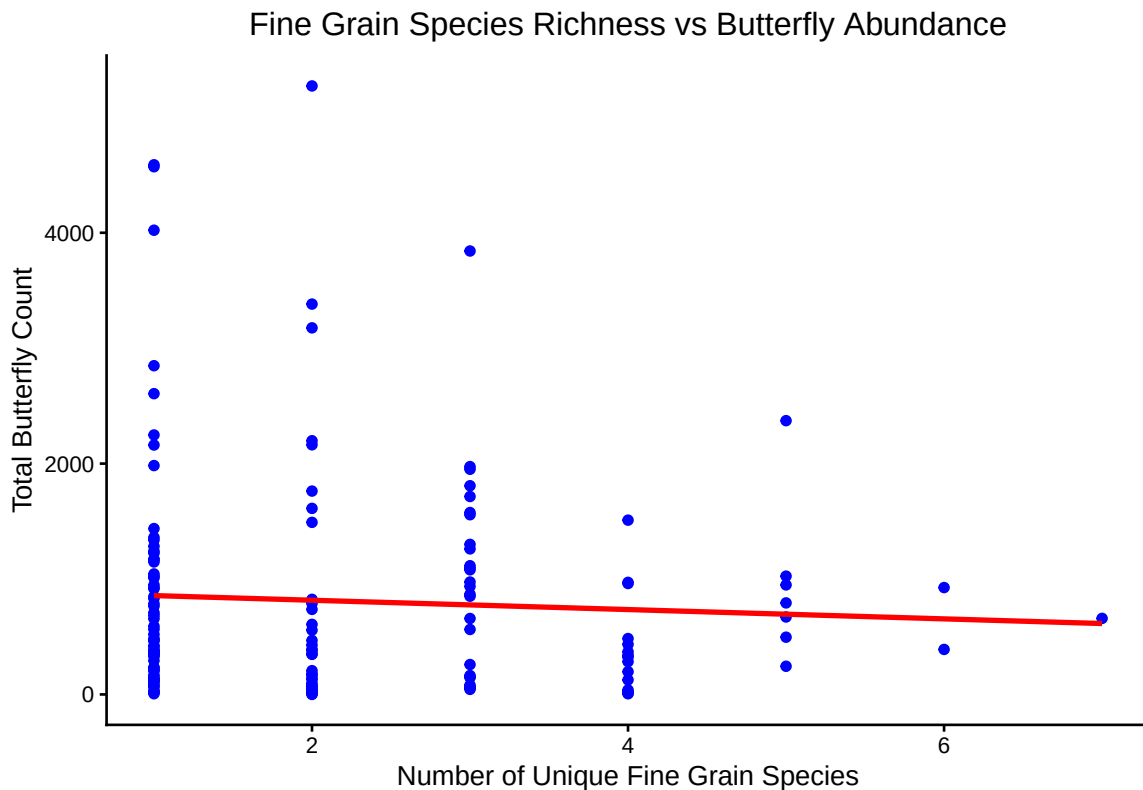
```
# Plot
ggplot(combined_dataset, aes(x = Unique_FG_Species, y = Total_Butterflies,
                             color = SITECODE)) +
  geom_point() +
  labs(
    title = "Fine Grain Species Richness vs Butterfly Abundance",
    x = "Number of Unique Fine Grain Species",
    y = "Total Butterfly Count"
  ) +
  theme_classic() +
  theme(plot.title = element_text(hjust = 0.5))
```



The scatter plot reveals a weak positive relationship between fine-grain species richness and butterfly abundance. Sites with low species richness consistently record low butterfly counts, whilst at higher richness values the butterfly counts become considerably more variable, ranging from very low to extremely high. Colouring points by SITECODE reveals that certain sites such as T05 and T10 consistently record higher butterfly counts regardless of fine grain species richness, suggesting site-level differences may be a more important driver of butterfly abundance than plant diversity alone.

c)

```
ggplot(combined_dataset, aes(x = Unique_FG_Species, y = Total_Butterflies)) +
  geom_point(color = "blue") +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  labs(
    title = "Fine Grain Species Richness vs Butterfly Abundance",
    x = "Number of Unique Fine Grain Species",
    y = "Total Butterfly Count"
  ) +
  theme_classic() +
  theme(plot.title = element_text(hjust = 0.5))
```



```
# Remove the outlier species

# Finding Quartiles
Q1 <- quantile(combined_dataset$Total_Butterflies, 0.25, na.rm = TRUE)
Q3 <- quantile(combined_dataset$Total_Butterflies, 0.75, na.rm = TRUE)

IQR_value <- IQR(combined_dataset$Total_Butterflies, na.rm = TRUE)

# Bounds for Outliers
lower_bound <- Q1 - 1.5 * IQR_value
upper_bound <- Q3 + 1.5 * IQR_value

# Removing Outliers
data_no_outlier <- combined_dataset %>%
  filter(
    Total_Butterflies >= lower_bound &
    Total_Butterflies <= upper_bound
  )

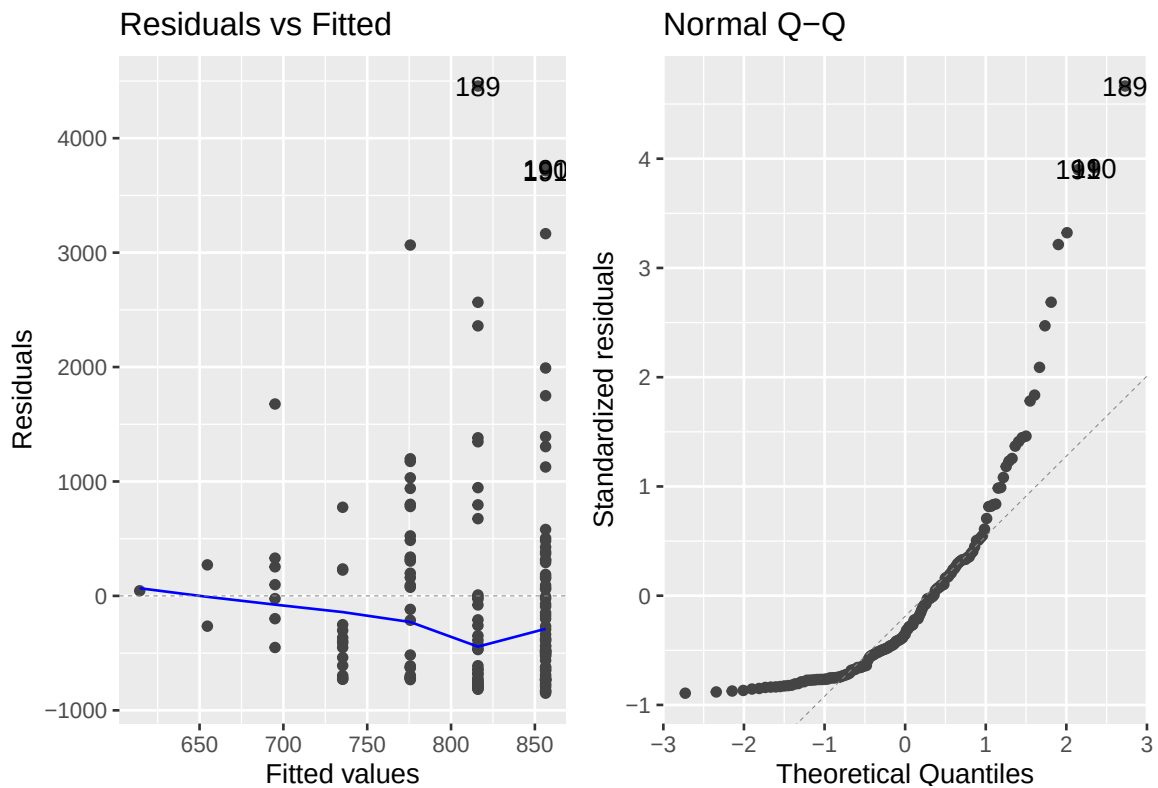
# Plot Without Outliers
ggplot(data_no_outlier, aes(x = Unique_FG_Species, y = Total_Butterflies)) +
  geom_point(color = "blue") +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
```

```
labs(
  title = "Relationship Without Extreme Outlier",
  x = "Number of Unique Fine Grain Species",
  y = "Total Butterfly Count"
) +
theme_classic() +
theme(plot.title = element_text(hjust = 0.5))
```

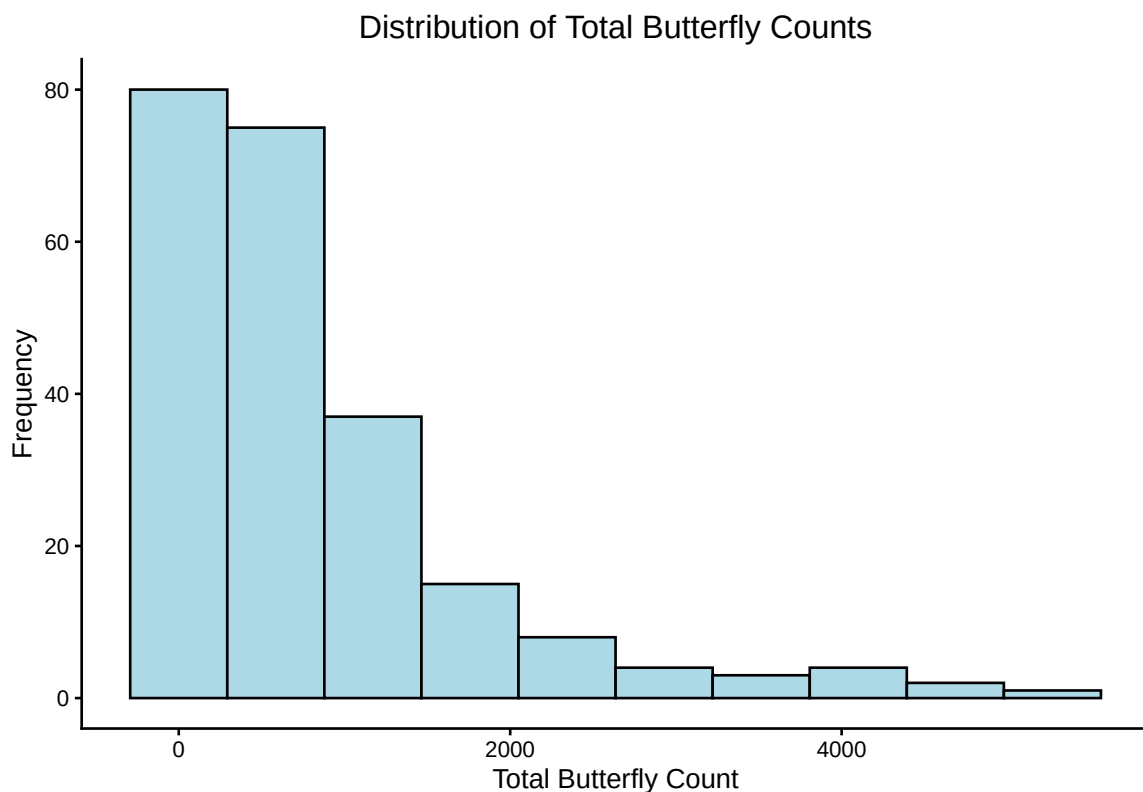


```
# Models (with and without outliers)
model <- lm(Total_Butterflies ~ Unique_FG_Species, data = combined_dataset)
model_no_outliers <- lm(Total_Butterflies ~ Unique_FG_Species, data = data_no_outlier)
r_squared <- summary(model)$r.squared
r_squared_no_outliers <- summary(model_no_outliers)$r.squared

# Further Info Plots on Overall Model
autoplot(model, which = 1:2)
```



```
# Histogram - Showing skewness and presence of extreme values
ggplot(combined_dataset, aes(x = Total_Butterflies)) +
  geom_histogram(fill = "lightblue", color = "black", bins = 10) +
  labs(title = "Distribution of Total Butterfly Counts",
       x = "Total Butterfly Count", y = "Frequency") +
  theme_classic() +
  theme(plot.title = element_text(hjust = 0.5))
```



```
# Table showing Gradient and Intercept of the Regression lines
table_model <- bind_rows(
  tidy(model) %>% mutate(Model = "With Outlier"),
```

```

tidy(model_no_outliers) %>% mutate(Model = "Without Outlier")
)

table_model <- table_model %>%
  mutate(term = case_when(
    term == "(Intercept)" ~ "Intercept",
    term == "Unique_FG_Species" ~ "Gradient",
    TRUE ~ term
  ))

kable(table_model, digits = 3, caption = "Linear Model Coefficients With and Without Outliers")

```

Table 2: Linear Model Coefficients With and Without Outliers

term	estimate	std.error	statistic	p.value	Model
Intercept	896.757	147.124	6.095	0.000	With Outlier
Gradient	-40.351	58.208	-0.693	0.489	With Outlier
Intercept	577.714	92.880	6.220	0.000	Without Outlier
Gradient	22.466	36.150	0.621	0.535	Without Outlier

The initial model suggests a slight negative relationship between fine-grain species richness and butterfly abundance, though this is not statistically significant (gradient = -40.35, $p = 0.489$). Interestingly, once outliers are removed the gradient actually flips direction (gradient = 22.47, $p = 0.535$), remaining non-significant. This instability in both the magnitude and direction of the gradient is concerning and suggests the apparent relationship is highly sensitive to a small number of extreme observations rather than reflecting a genuine underlying trend in the data.

The diagnostic plots raise further concerns about the validity of the model. The Q-Q plot shows that residuals deviate substantially from normality, particularly in the upper tail, whilst the residuals vs fitted plot reveals clear heteroscedasticity, with variance increasing at higher fitted values. The histogram of total butterfly counts confirms the response variable is heavily right-skewed, which partly explains these violations. Together with a low R^2 of 0.003, these diagnostics suggest that simple linear regression is not well-suited to this data. Butterfly abundance is likely influenced by more complex ecological factors such as site-level characteristics, weather conditions, or habitat quality that a single predictor model cannot adequately capture.

d)

```
# Remove rows with missing fine-grain data before fitting site models
data_fg <- combined_dataset %>%
  filter(!is.na(Unique_FG_Species))

sites_fg <- unique(data_fg$SITECODE)

for (site in sites_fg) {

  site_data <- data_fg %>% filter(SITECODE == site)

  # Skip if not enough data
  if (nrow(site_data) < 2) {
    cat(paste("\nSite", site, "- not enough data, so skipping\n"))
    next
  }

  model_site <- lm(Total_Butterflies ~ Unique_FG_Species, data = site_data)

  # Scatter + regression line per site
  print(
    ggplot(site_data, aes(x = Unique_FG_Species, y = Total_Butterflies)) +
      geom_point(color = "blue") + geom_smooth(method = "lm", color = "red", se = FALSE) +
      labs(title = paste("Site", site, ": Fine Grain Species Richness vs Butterfly Abundance"),
           x = "Number of Unique Fine Grain Species",
           y = "Total Butterfly Count"
      ) + theme_classic() + theme(plot.title = element_text(hjust = 0.5))
  )

  # Coefficient table per site
  print(
    kable(tidy(model_site), digits = 3,
           caption = paste("Linear Model Coefficients for Site", site))
  )
}
```

Site T01 : Fine Grain Species Richness vs Butterfly Abundance

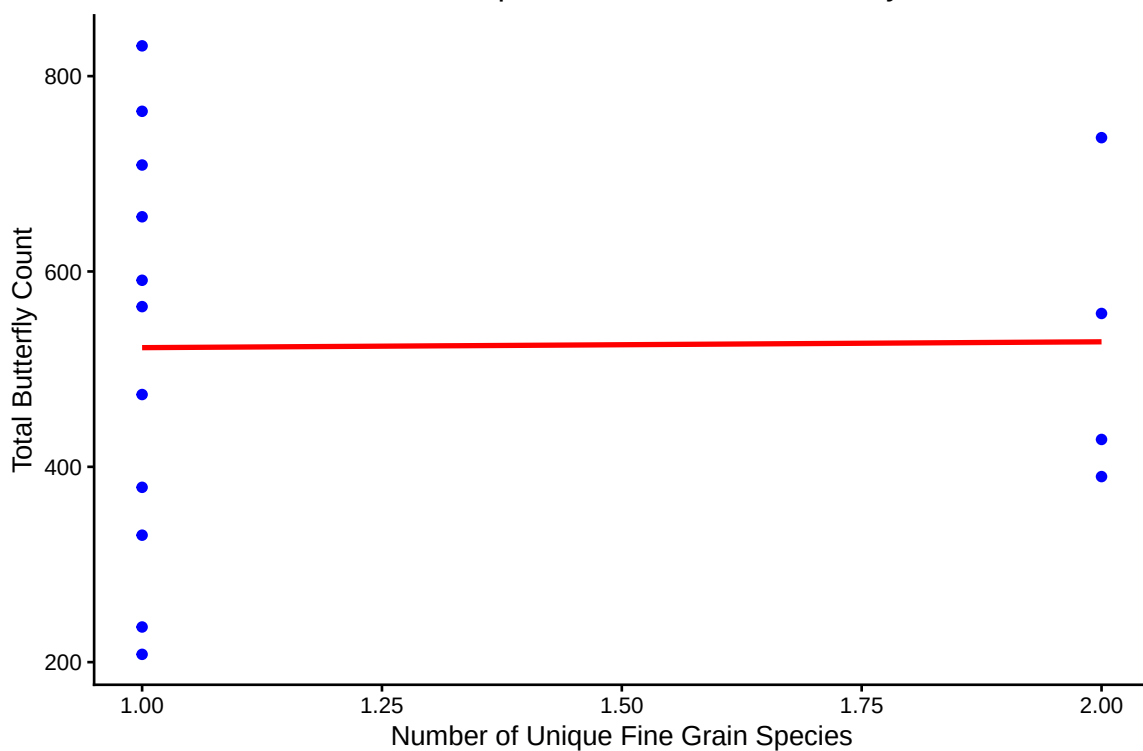


Table 3: Linear Model Coefficients for Site T01

term	estimate	std.error	statistic	p.value
(Intercept)	516	157.731	3.271	0.006
Unique_FG_Species	6	117.566	0.051	0.960

Site T02 : Fine Grain Species Richness vs Butterfly Abundance

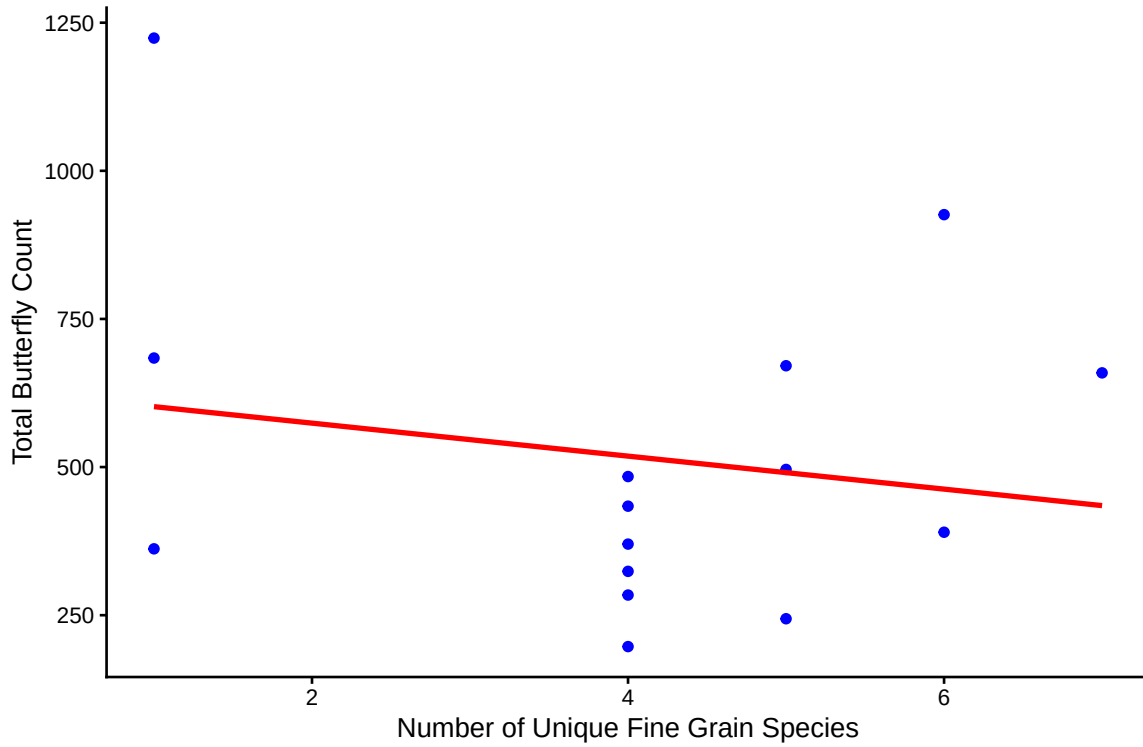


Table 4: Linear Model Coefficients for Site T02

term	estimate	std.error	statistic	p.value
(Intercept)	629.814	183.195	3.438	0.004
Unique_FG_Species	-27.839	41.309	-0.674	0.512

Site T03 : Fine Grain Species Richness vs Butterfly Abundance

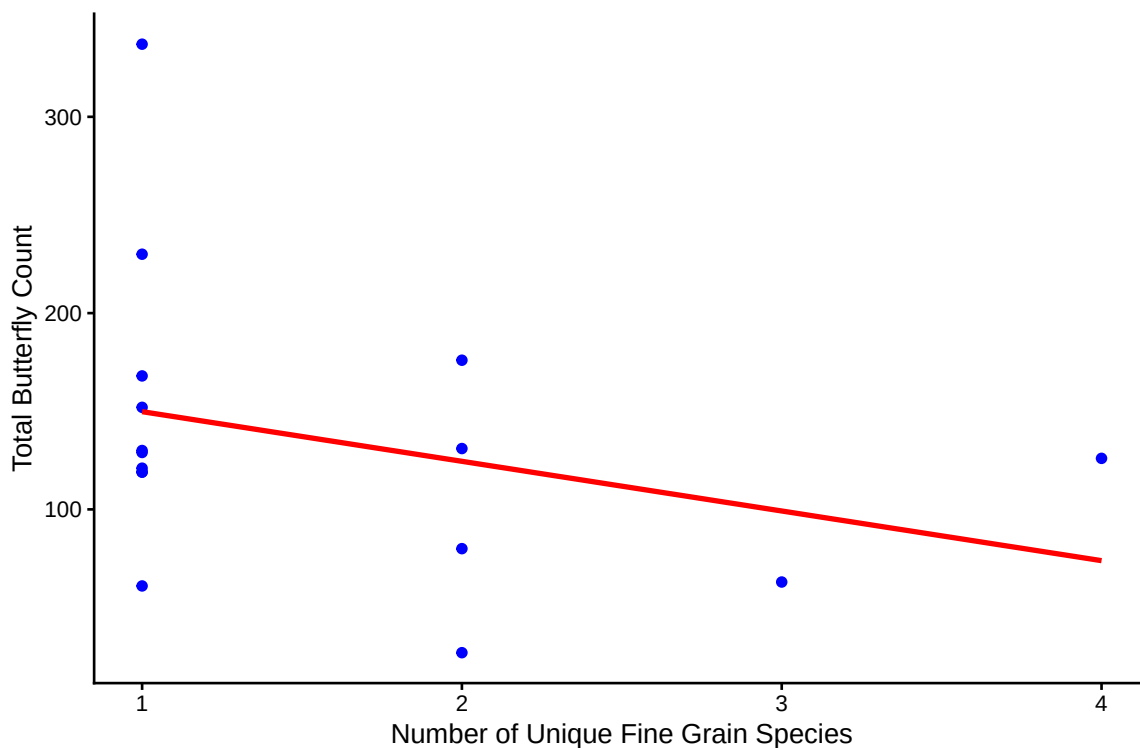


Table 5: Linear Model Coefficients for Site T03

term	estimate	std.error	statistic	p.value
(Intercept)	175.099	36.916	4.743	0.000
Unique_FG_Species	-25.304	20.677	-1.224	0.241

Site T04 : Fine Grain Species Richness vs Butterfly Abundance

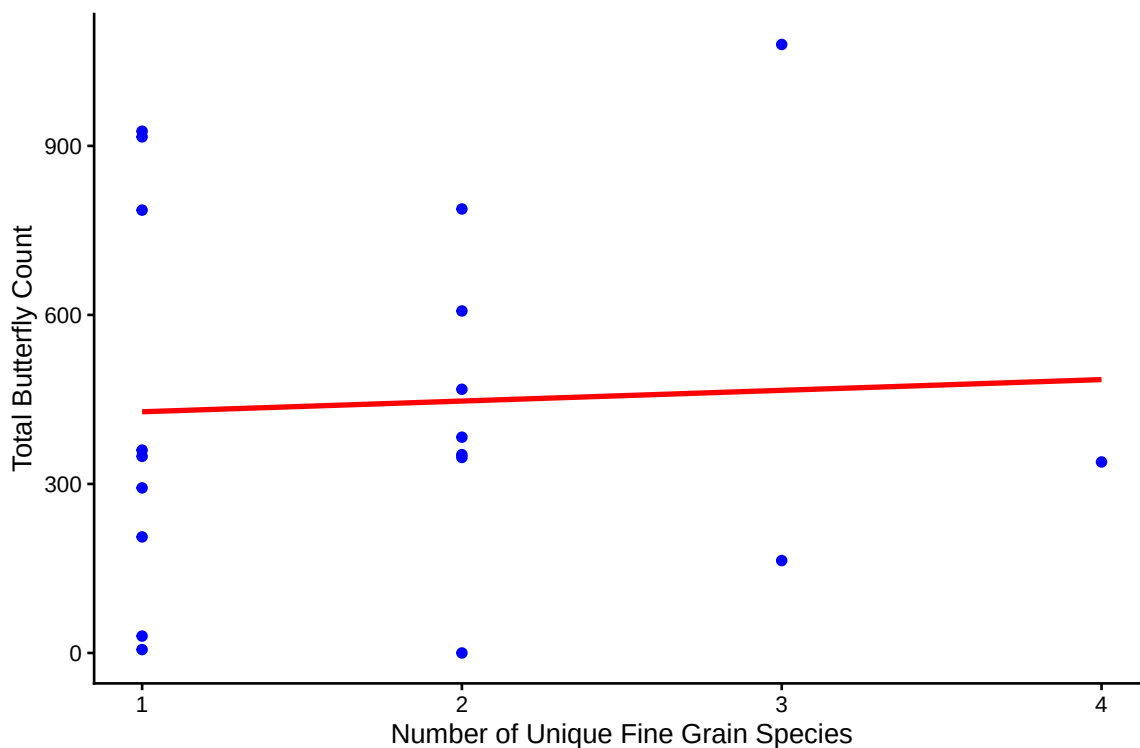


Table 6: Linear Model Coefficients for Site T04

term	estimate	std.error	statistic	p.value
(Intercept)	409.038	173.961	2.351	0.031
Unique_FG_Species	19.038	89.991	0.212	0.835

Site T05 : Fine Grain Species Richness vs Butterfly Abundance

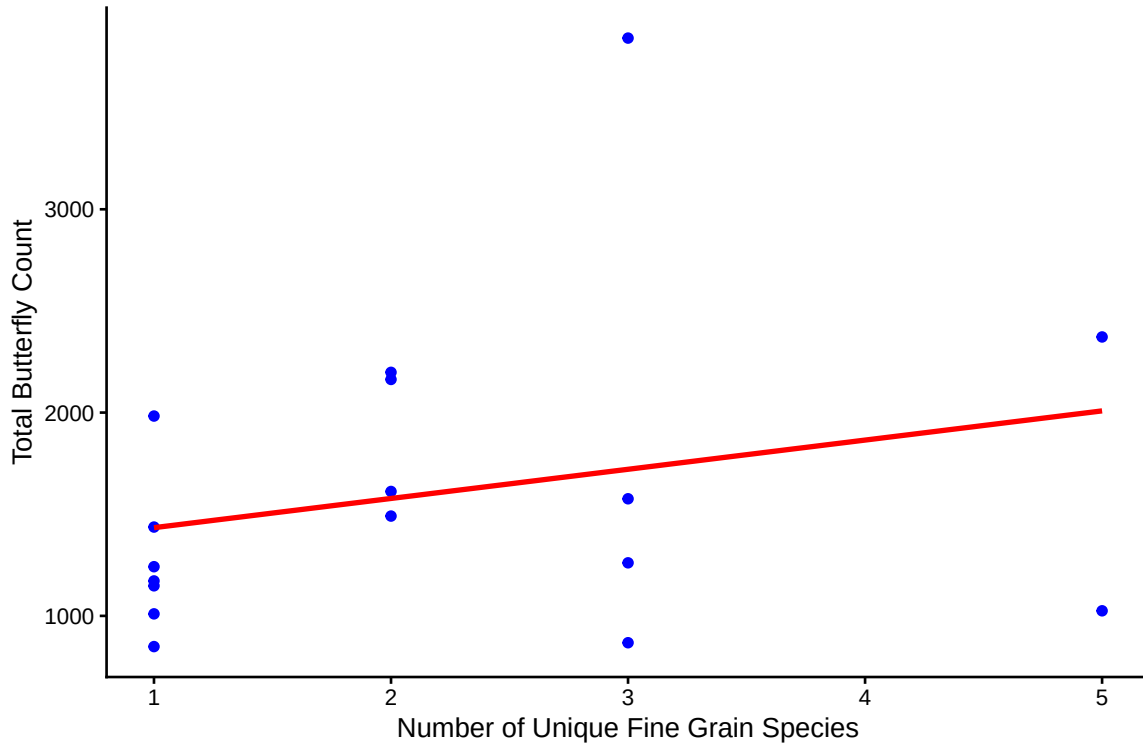


Table 7: Linear Model Coefficients for Site T05

term	estimate	std.error	statistic	p.value
(Intercept)	1290.343	353.111	3.654	0.002
Unique_FG_Species	143.599	139.451	1.030	0.319

Site T06 : Fine Grain Species Richness vs Butterfly Abundance

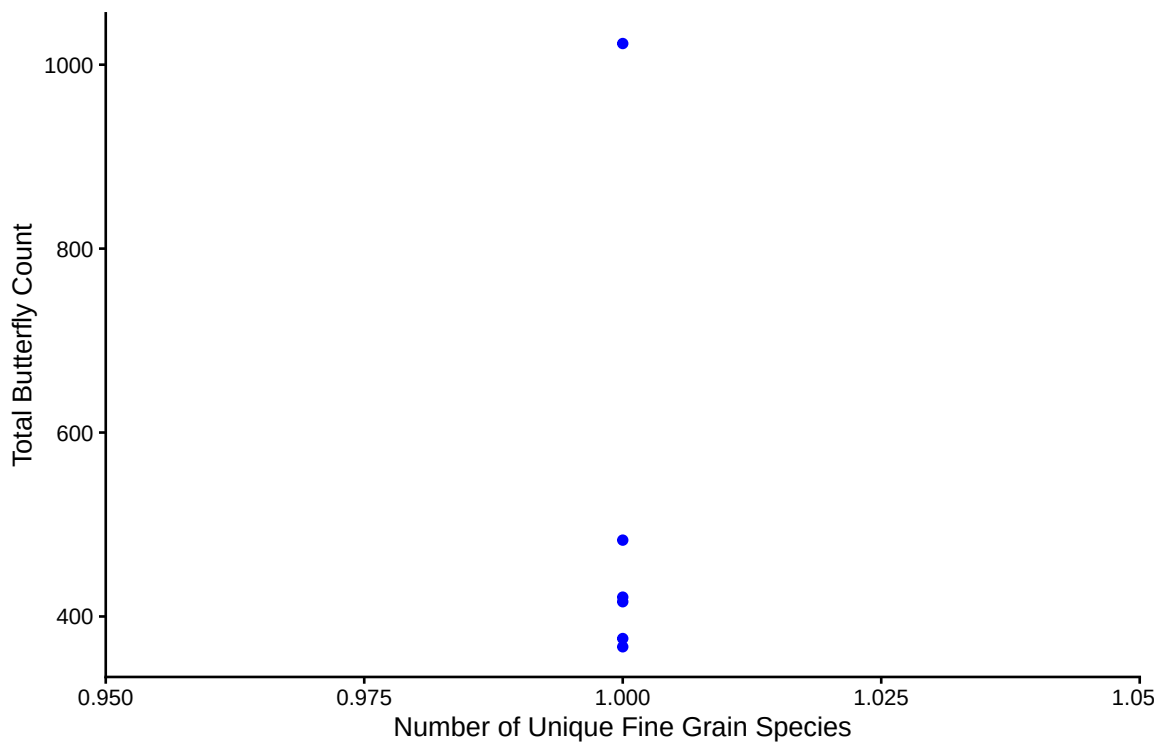


Table 8: Linear Model Coefficients for Site T06

term	estimate	std.error	statistic	p.value
(Intercept)	514.333	103.11	4.988	0.004
Unique_FG_Species	NA	NA	NA	NA

Site T07 : Fine Grain Species Richness vs Butterfly Abundance

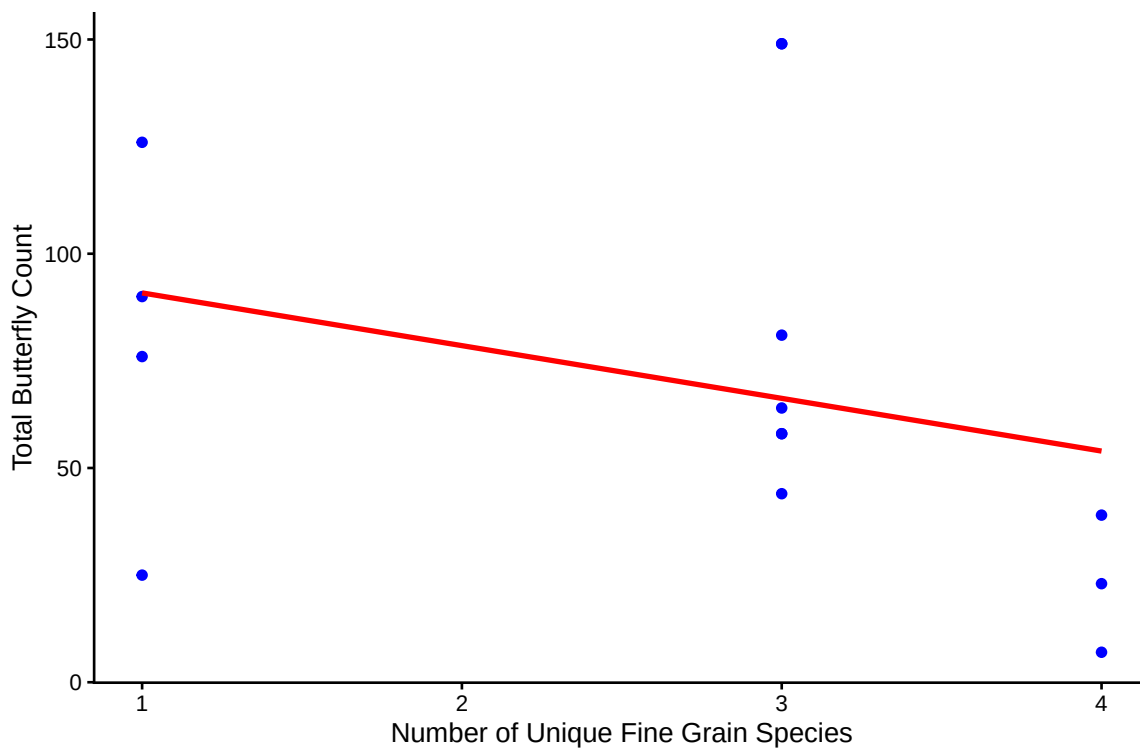


Table 9: Linear Model Coefficients for Site T07

term	estimate	std.error	statistic	p.value
(Intercept)	103.158	30.717	3.358	0.006
Unique_FG_Species	-12.303	10.717	-1.148	0.273

Site T08 : Fine Grain Species Richness vs Butterfly Abundance

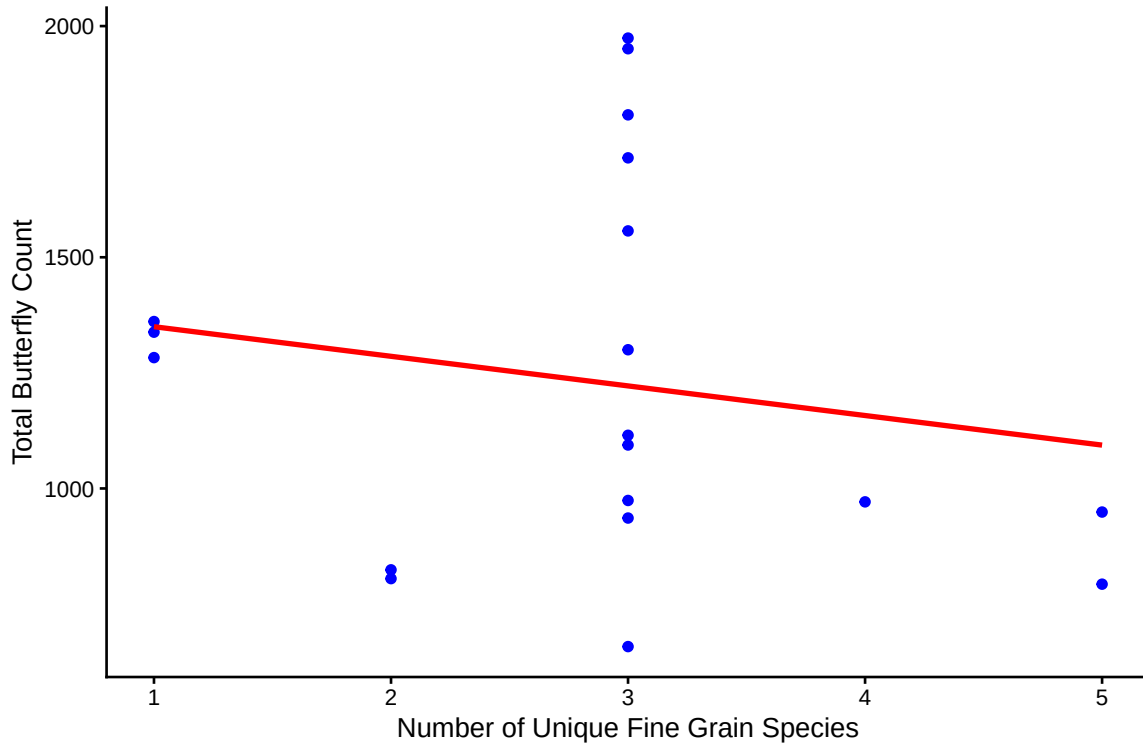


Table 10: Linear Model Coefficients for Site T08

term	estimate	std.error	statistic	p.value
(Intercept)	1413.869	264.265	5.350	0.000
Unique_FG_Species	-64.028	86.828	-0.737	0.471

Site T09 : Fine Grain Species Richness vs Butterfly Abundance

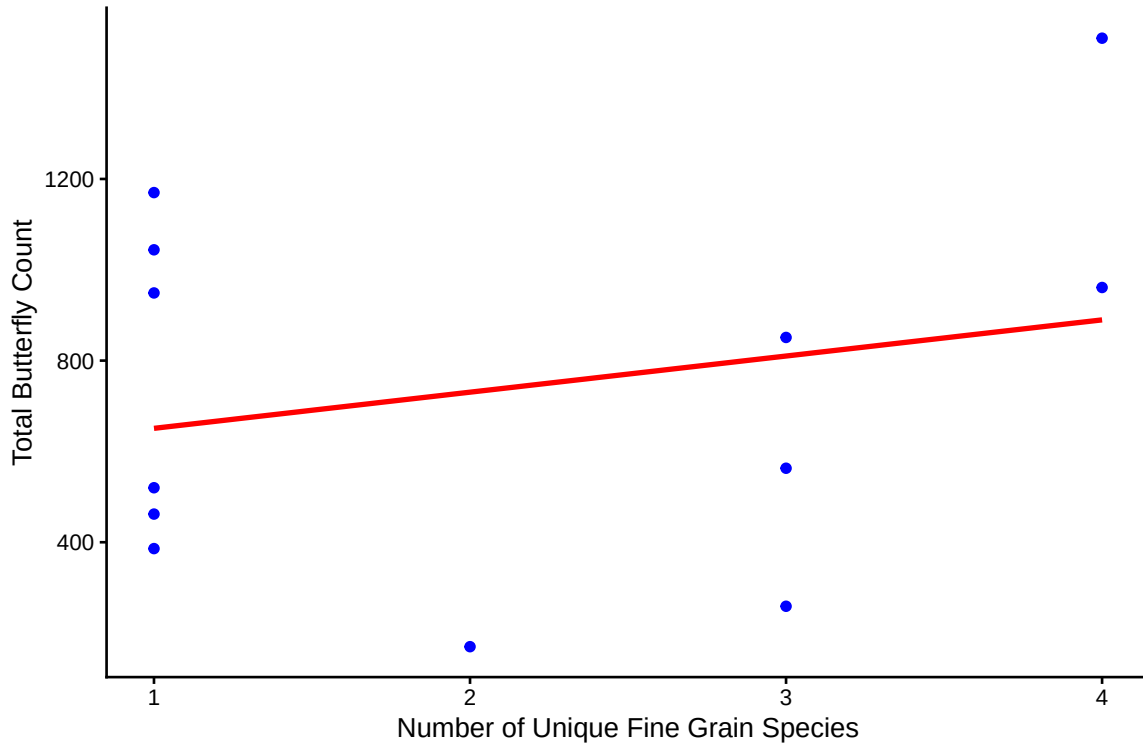


Table 11: Linear Model Coefficients for Site T09

term	estimate	std.error	statistic	p.value
(Intercept)	571.207	240.827	2.372	0.039
Unique_FG_Species	79.621	100.432	0.793	0.446

Site T10 : Fine Grain Species Richness vs Butterfly Abundance

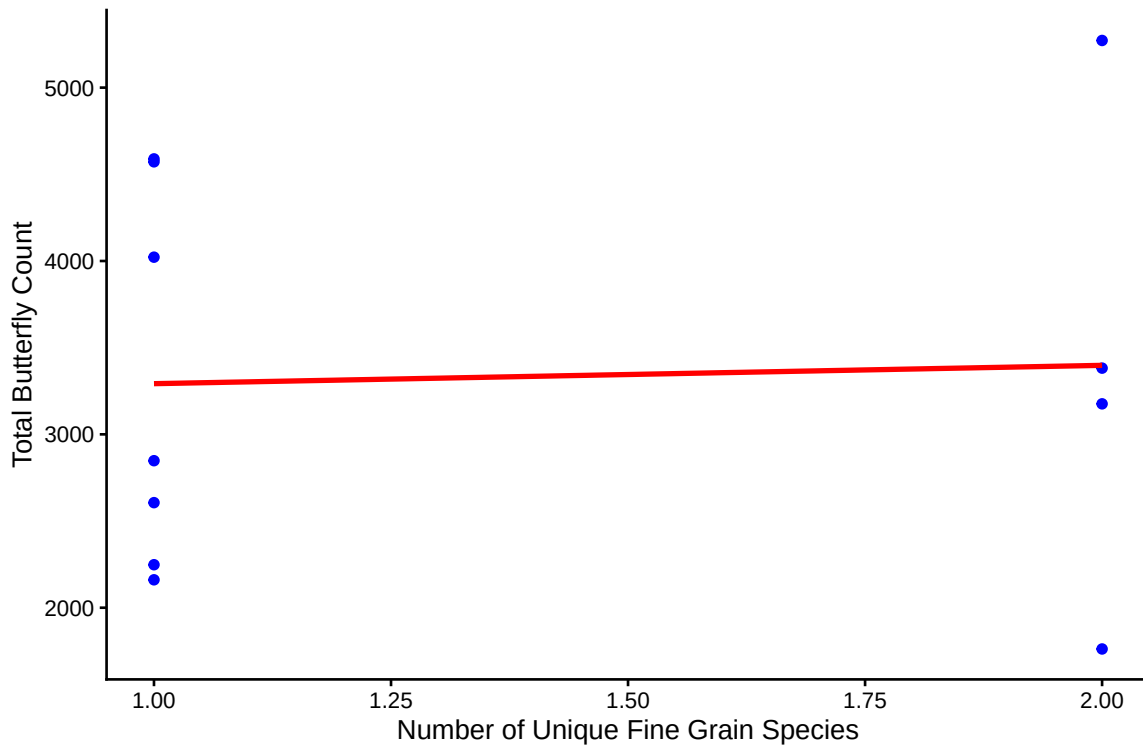


Table 12: Linear Model Coefficients for Site T10

term	estimate	std.error	statistic	p.value
(Intercept)	3186.571	1094.629	2.911	0.017
Unique_FG_Species	105.714	757.006	0.140	0.892

Site T11 : Fine Grain Species Richness vs Butterfly Abundance

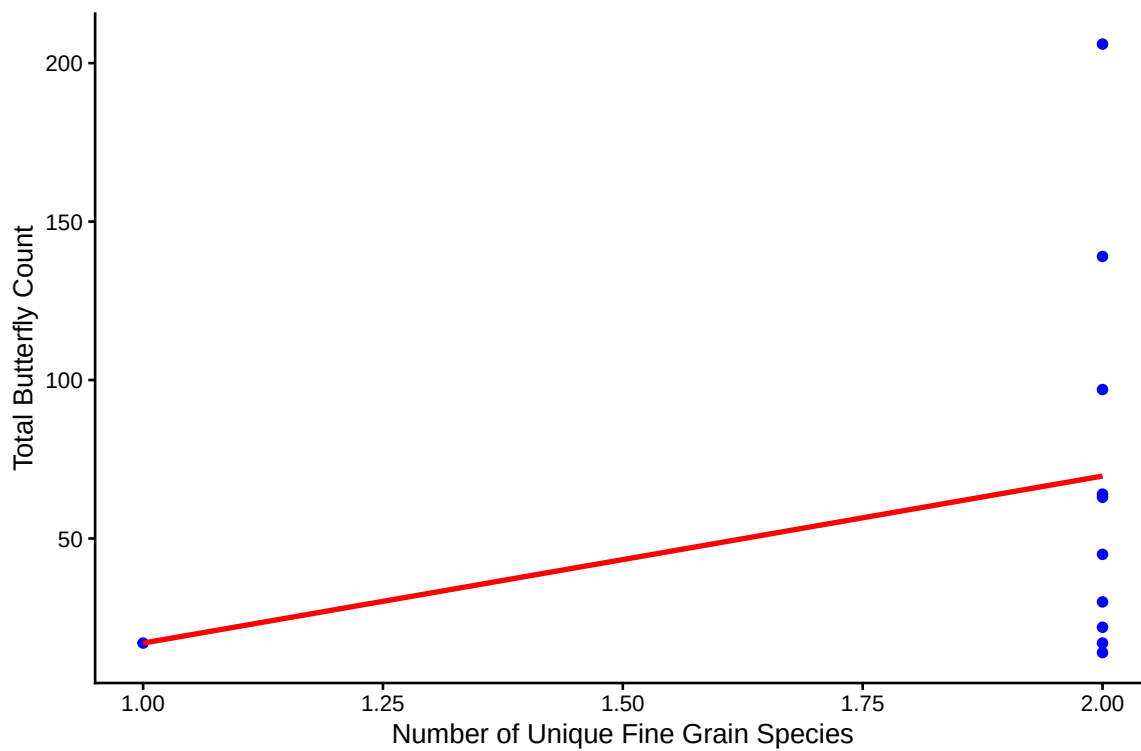


Table 13: Linear Model Coefficients for Site T11

term	estimate	std.error	statistic	p.value
(Intercept)	-35.7	125.446	-0.285	0.782
Unique_FG_Species	52.7	64.977	0.811	0.438

Site T12 : Fine Grain Species Richness vs Butterfly Abundance

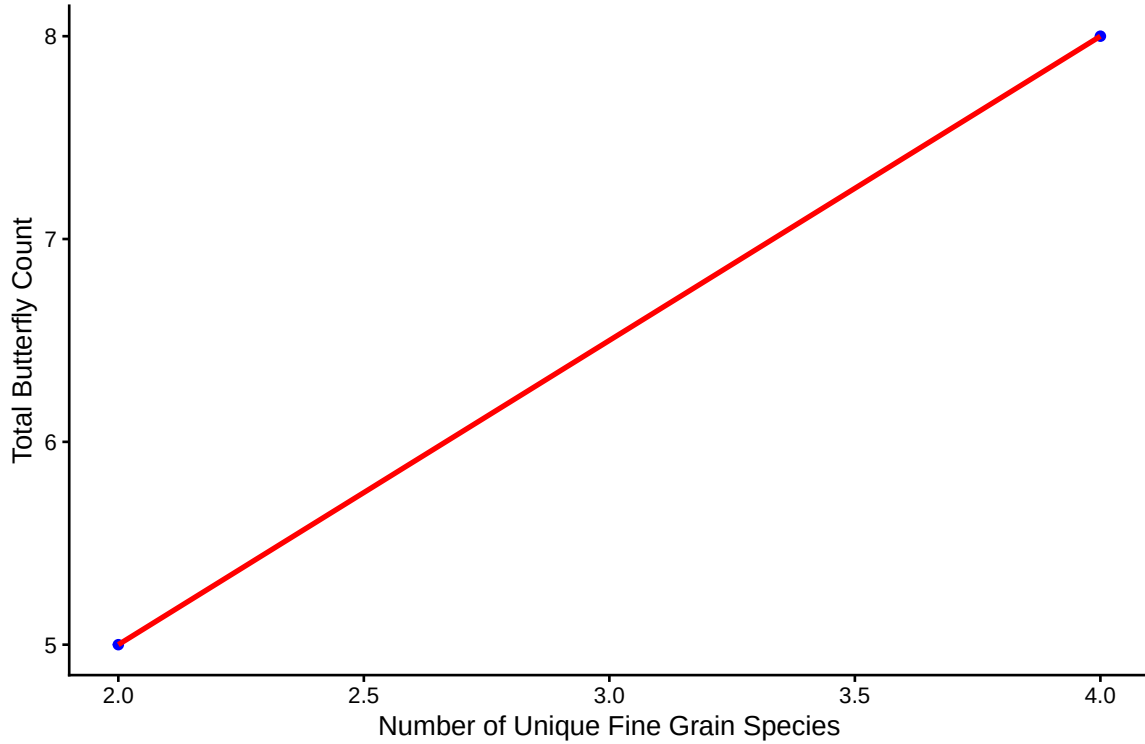


Table 14: Linear Model Coefficients for Site T12

term	estimate	std.error	statistic	p.value
(Intercept)	2.0	NaN	NaN	NaN
Unique_FG_Species	1.5	NaN	NaN	NaN

Site-specific models reveal little to no consistent relationship between fine-grain species richness and butterfly abundance across the network. Most slopes are non-significant. Site T06 had no variation in fine-grain species richness (all observations recorded exactly one species), making the slope unidentifiable. T12 had only two observations, producing a perfect fit with undefined standard errors and no meaningful inference possible. The site-specific regression lines deviate considerably from the global model in several panels, confirming the global model poorly represents individual sites. This is consistent with part c), where the apparent relationship appears driven by outliers and heterogeneity between sites rather than a genuine network-wide trend.